CHARACTERISATIONS OF A DOMINANT MORPHISM OF FINITE DEGREE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finiteness of a finite type extension of domains detected by the field extension, up to localisation. Algebra, Lemma \ref{algebra-lemma-finite-extension-finitely-generated-fractions} and the surrounding discussion, used at proof lines 13418-13445. Translates between the field and ring statements.
- Generic point conditions for finite type maps of integral schemes. Lemma \ref{lemma-generically-finite} and Lemma \ref{lemma-quasi-finiteness-over-generic-point}, used at proof lines 13446-13459. Give the pointwise formulations.

Source reference. Original lemma 02NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3496–3496. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01RM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2307–2307. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02FX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3208–3208. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00GS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 618–618. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BAI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14211–14211. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let X, Y be integral schemes. Let $f : X \to Y$ be locally of finite type. Assume f is dominant. The following are equivalent:*

1. *the extension $R(Y) \subset R(X)$ has transcendence degree 0,*
2. *the extension $R(Y) \subset R(X)$ is finite,*
3. *there exist nonempty affine opens $U \subset X$ and $V \subset Y$ such that $f(U) \subset V$ and $f|_U : U \to V$ is finite, and*
4. *the generic point of X is the only point of X mapping to the generic point of Y.*

If f is separated or if f is quasi-compact, then these are also equivalent to

5. *there exists a nonempty affine open $V \subset Y$ such that $f^{-1}(V) \to V$ is finite.*

Proof. Choose any affine opens $\mathrm{Spec}(A) = U \subset X$ and $\mathrm{Spec}(R) = V \subset Y$ such that $f(U) \subset V$. Then R and A are domains by definition. The ring map $R \to A$ is of finite type (Lemma 01T2). By Lemma 01RM the generic point of X maps to the generic point of Y hence $R \to A$ is injective. Let $K = R(Y)$ be the fraction field of R and $L = R(X)$ the fraction field of A. Then L/K is a finitely generated field extension. Hence we see that (1) is equivalent to (2).

Suppose (2) holds. Let $x_1, \ldots, x_n \in A$ be generators of A over R. By assumption there exist nonzero polynomials $P_i(X) \in R[X]$ such that $P_i(x_i) = 0$. Let $f_i \in R$ be the leading coefficient of P_i. Then we conclude that $R_{f_1 \ldots f_n} \to A_{f_1 \ldots f_n}$ is finite, i.e., (3) holds. Note that (3) implies (2). So now we see that (1), (2) and (3) are all equivalent.

Let η be the generic point of X, and let $\eta' \in Y$ be the generic point of Y. Assume (4). Then $\dim_\eta(X_{\eta'}) = 0$ and we see that $R(X) = \kappa(\eta)$ has transcendence degree 0 over $R(Y) = \kappa(\eta')$ by Lemma 02FX. In other words (1) holds. Assume the equivalent conditions (1), (2) and (3). Suppose that $x \in X$ is a point mapping to η'. As x is a specialization of η, this gives inclusions $R(Y) \subset \mathcal{O}_{X,x} \subset R(X)$, which implies $\mathcal{O}_{X,x}$ is a field, see Algebra, Lemma 00GS. Hence $x = \eta$. Thus we see that (1) – (4) are all equivalent.

It is clear that (5) implies (3) with no additional assumptions on f. What remains is to prove that if f is either separated or quasi-compact, then the equivalent conditions (1) – (4) imply (5). This follows from Lemma 0BAI. □